

DATA 202 — Quiz 1

Week 2 — DataFrame Basics & Visual Encodings

Name: _____

Date: _____

The Dataset: Product Sales

`sales` contains 50 simulated products. Each row describes one product's performance: how much it sold, how it was advertised, and how profitable it was.

Columns: Product ID, Sales, Units Sold, Profit, Advertising Spend (numerical)
· Category, Season, Supplier (categorical)

First few rows:

Product ID	Sales	Units Sold	Profit	Advertising Spend	Category	Season	Supplier
P014	36921	1391	8582	5925	Electronics	Spring	Supplier B
P020	33496	1181	-4775	8432	Furniture	Summer	Supplier A
P036	32560	968	9691	4040	Clothing	Winter	Supplier B
P041	58147	1155	7637	3282	Furniture	Fall	Supplier B
P047	82811	698	8838	4530	Clothing	Summer	Supplier C

Question 1 (SLO 02A)

For each command below, identify whether it **Accesses**, **Adds**, or **Deletes** data, and whether it operates on a **Row** or a **Column**.

#	Command	Access / Add / Delete?	Row(s) or Column?
1	<code>sales["Units Sold"]</code>		
2	<code>sales["Margin"] = sales["Profit"] / sales["Sales"]</code>		
3	<code>sales = pd.concat([sales, pd.DataFrame([new_row])], ignore_index=True)</code>		
4	<code>sales = sales[sales["Category"] != "Electronics"]</code>		
5	<code>sales = sales.drop(columns=["Supplier"])</code>		

Question 2 (SLO 02B)

Complete the code below to select all products in the **Furniture** category that lost money (**Profit below \$0**), sorted by Profit from lowest to highest.

Remember: the filter condition still goes inside the square brackets and must reference `sales` again,

e.g. `sales[sales["Column"] ...]`.

```
result = sales[
    _____
]
result = result.sort_values("_____", ascending=_____)
print(len(result))
```

Question 3 (SLO 02C)

Available visual channels: `x`, `y`, `color`, `size`, `symbol`, `facet_col`.

Since this is a paper quiz, `color` isn't something you can sketch in pencil — the code below uses `symbol` instead, which maps a categorical column to different *marker shapes* (e.g. $\times$, $\circ$, $\triangle$) rather than colors.

For the code below, fill in the table: for each visual channel it uses, name the column mapped to it and say whether that column is Numerical or Categorical.

```
px.scatter(sales, x="Advertising Spend", y="Sales",
           symbol="Season", size="Units Sold",
           title="Advertising Spend vs. Sales")
```

Visual Channel	Column Used	Data Type (Numerical / Categorical)
x-axis		
y-axis		
symbol		
size		

In the space below, sketch what you expect this chart to look like. Label your axes. You don't need exact data — just show the general pattern (point cloud, marker shapes ($\times/\circ/\triangle$) for groups, varying point sizes).